

Tetraphenylethylene-Based Emissive Supramolecular Metallacages Assembled by Terpyridine Ligands

Meng Li^{1†}, Shan Jiang^{1†}, Zhe Zhang², Xin-Qi Hao³, Xin Jiang¹, Hao Yu¹, Pingshan Wang², Bin Xu^{1*}, Ming Wang^{1*} & Wenjing Tian¹

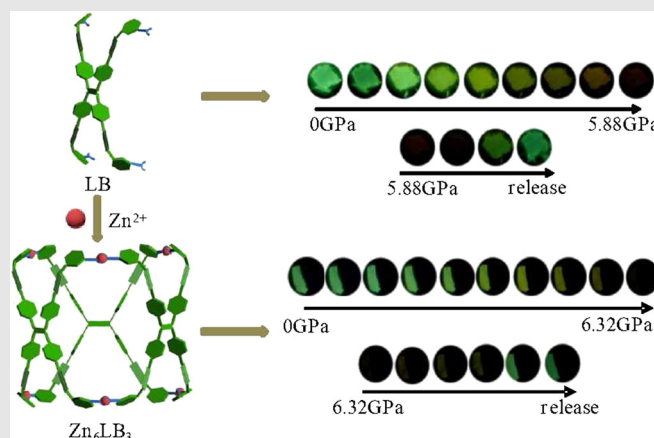
¹State Key Laboratory of Supramolecular Structure and Materials, College of Chemistry, Jilin University, Changchun, Jilin 130012, ²Institute of Environmental Research at Greater Bay Area, Key Laboratory for Water Quality and Conservation of the Pearl River Delta, Ministry of Education, Guangzhou University, Guangzhou 510006, ³College of Chemistry and Green Catalysis Center, Zhengzhou University, Zhengzhou 450002.

*Corresponding authors: mingwang358@jlu.edu.cn; xubin@jlu.edu.cn; [†]M. Li and S. Jiang contributed equally to this work.

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We report the preparation and emission properties of tetraphenylethylene (TPE)-based metallacages with aggregation-induced emission (AIE) activities through coordination-driven self-assembly. Two supramolecular cages, $[\text{Zn}_6\text{LA}_3]$ and $[\text{Zn}_6\text{LB}_3]$, were assembled via TPE-decorated terpyridine (tpy) ligands, LA and LB, respectively, with $\text{Zn}(\text{II})$ ions. We performed a subtle change by introducing extra alkyne connectivity into LB to increase the degree of conjugation and geometric constraint, compared with LA. As a result, we obtained a highly emissive cage, $[\text{Zn}_6\text{LB}_3]$, even in a dilute solution. At a low temperature, the intramolecular rotation of TPE was further restricted, thus, resulting in a significant increase in fluorescence. Through mixing LA and LB, we obtained a series of hybrid cages, which also indicated that the emission was enhanced with highly abundant LB in the cages. Further, we studied the emission behaviors of the ligands and cages in solid state under external pressure. Upon gradual increase of the external pressure, the luminescence of $[\text{Zn}_6\text{LB}_3]$ increased initially, due to further rotation restriction, which was followed by quenching under

6.32 GPa, owing to the tight packing of the supramolecules. The subsequent release of the pressure resulted in cage recovery of the emission.



Keywords: self-assembly, terpyridine, tetraphenylethylene, AIE

Introduction

In the past two decades, aggregation-induced emission (AIE) studies have made significant progress in light-emitting materials,^{1–10} after Tang and co-workers¹¹ first coined the concept of AIE in 2001. As a breakthrough study, the AIE phenomenon is completely opposite to the aggregation-caused quenching (ACQ) effect of conventional organic fluorophores. It is well documented that the aggregates of AIE-active molecules are able to block the thermal decay of nonluminescent substances in solution, leading to higher fluorescence efficiency than the single AIE molecule.^{12–19} Tetraphenylethylene (TPE) has been explored extensively as one of the iconic AIE fluorophores in terms of its fundamental emissive mechanism and its intriguing application.²⁰ Within the TPE, free rotation of the phenyl rings led to fast nonradiative decay in a dilute solution. Among the TPE aggregates, due to the effect of restricting intramolecular motion, there is a hindrance of the path of nonradiative transitions and thus, induces fluorescence emission through radiation transitions.^{21–28} Specifically, owing to the largely twisted structure and the steric hindrance by phenyl rings, TPE derivatives often show distinct optical response under external stimuli, such as thermal energy and pressure applications. Based on such a distinct optical behavior, a series of TPE-related motifs have been introduced into supramolecular systems to target a variety of applications, including fluorescence sensing,^{1,29,30} imaging,^{18,31} recognition,³² catalysis,^{33,34} and others.

Thanks to the facile synthesis and rigid geometry, TPE and its derivatives exhibited remarkable compatibility for most of metalla-supramolecular systems. Currently, a series of discrete two-dimensional (2D) macrocycles, three-dimensional (3D) cages, and nanomaterials with well-defined sizes and shapes were constructed by coordination-driven self-assembly.^{35–50} In this field, Stang and co-workers⁵¹ reported the first TPE-based supramolecular cage, which brought about the recent flourishing of AIE-active metalla-supramolecular cages with tunable properties.^{52–54} Among most of these structures, the TPE moieties were either attached or embedded in the supramolecules, and their emissions still followed the classic AIE principle. Accordingly, high fluorescent efficiency was mainly observed in their aggregate state, in which the rotations of the benzene rings were restricted effectively.⁵⁵ Nonetheless, in the solution state, as well as the isolated supramolecular cage scenario, these nonaggregate supramolecules were still nonemissive or weakly luminescent,^{47,48,50,54,56–60} which severely limited the application prospect of the TPE molecule.

Herein, we intended to design and assemble discrete fluorescent supramolecules using the TPE motif with our goal of overcoming the limitation mentioned above to achieve high fluorescent efficiency in solution by

extending the conjugation and rigidity of its building blocks. Accordingly, we report the self-assembly of two 2,2':6'',2''-terpyridine (tpy)-based metallacages with tunable luminescence properties. The final discrete M_6L_3 cages were formed by the coordination between TPE-based tpy ligands and Zn(II). The structures of the supramolecules were characterized fully by NMR and mass spectrometry, and the optical properties were characterized by UV-vis, fluorescence, low-temperature emission, and high-pressure emission spectroscopy.

Experimental Method

Materials and methods

All reagents were commercially available from Ark Pharm Inc. (Beijing, China) and Aladdin Bio-Chem Technology Co. Ltd (Shanghai, China) and were used without further purification. Compounds **2**,⁴⁹ **3**,⁵⁰ **7**,⁶¹ and **8**⁶² were synthesized according to the reported methods. NMR spectra data were recorded on AVANCE 500-MHz (Bruker, Beijing, China) and 600-MHz NMR spectrometer (Bruker) in deuterated chloroform ($CDCl_3$) and deuterated acetonitrile (CD_3CN) with tetramethylsilane (TMS) as standard. Electrospray ionization mass spectrometry (ESI-MS) and traveling-wave ion mobility mass spectrometry (TWIM-MS) were recorded with a SYNAPT G2 tandem mass spectrometer (Waters, Milford, MA, United States). The UV-vis spectra were recorded with a UV2550 spectrophotometer (Shimadzu, Shanghai, China). The emission spectra were measured on a Shimadzu RF-5301 PC spectrometer (CCD) and Maya2000Pro optical fiber spectrophotometer (Ocean Insight, Shanghai, China). The solution-state quantum yields were determined by the FLS920 steady- and transient-state fluorescence spectrometer (Edinburgh Instruments Ltd., Livingston, United Kingdom). The high-pressure fluorescence spectra under hydrostatic conditions were measured using a fluorescence microscope (IX71, 50, NA = 0.5; Olympus, Beijing, China) equipped with a spectrometer (Horiba Jobin Yvon iHR320; Bensheim, Germany). The light source was a mercury lamp with an excitation wavelength of 365 nm.

Synthesis of ligand **LA**

Compound **3** (0.91 g, 1.872 mmol), compound **8** (0.3 g, 0.36 mmol), $Pd(PPh_3)_2Cl_2$ (66 mg, 0.1 mmol), and Na_2CO_3 (1.0 g) were added into a Schlenk flask and degassed under nitrogen for three times. Next, 10 mL H_2O and 30 mL THF were injected and stirred at 70 °C for 4 days. After cooling to room temperature, the reactants were extracted with $CHCl_3$, and the crude product was purified by flash column chromatography with $CHCl_3$: ethanol (50:1, v/v) to obtain the ligand **LA** (0.47 g, 67% yield) as a yellow solid. 1H NMR conditions: (500 MHz,

CDCl_3 , 300 K); characterization, δ (ppm): 8.73 (s, 8H, $\text{tpy-H}^{3,5'}$), 8.66 (m, 16H, $\text{tpy-H}^{6,6''}$ and $\text{tpy-H}^{3,3''}$), 7.89–7.83 (m, 8H, $\text{tpy-H}^{4,4''}$), 7.80 (d, $J = 2.3$ Hz, 4H, H^b), 7.61 (d, $J = 8.2$ Hz, 4H, H^b), 7.47 (d, $J = 8.0$ Hz, 8H, H^d), 7.31 (d, $J = 6.3$ Hz, 8H, $\text{tpy-H}^{5,5''}$), 7.24 (d, $J = 8.2$ Hz, 8H, H^e), 7.04 (d, $J = 8.6$ Hz, 4H, H^c), 4.05 (t, $J = 6.3$ Hz, 8H, H^f), 1.74 (t, $J = 7.5$ Hz, 8H), 1.46–1.38 (m, 8H), 1.17 (m, 16H), 0.75 (t, $J = 7.1$ Hz, 12H). ^{13}C NMR conditions: (125 MHz, CDCl_3 , 300 K); characterization, δ (ppm): 156.57, 155.77, 155.08, 149.10, 148.49, 142.64, 140.21, 138.28, 136.63, 133.50, 132.03, 129.10, 128.57, 128.36, 126.14, 123.46, 121.98, 121.17, 112.57, 68.68, 31.56, 29.19, 25.81, 22.38, 13.94. Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS) characterization, (m/z): Calcd for $[\text{C}_{134}\text{H}_{120}\text{N}_{12}\text{O}_4\text{-e}]^+$ 1960.96. Found: 1960.95.

Synthesis of ligand LB

Compound **5** (0.14 g, 0.2 mmol), compound **7** (0.5 g, 1.2 mmol), $\text{Pd}(\text{PPh}_3)_4$ (46 mg, 0.04 mmol), CuI (6.0 mg, 0.032 mmol) were added into a Schlenk flask. After degassing under nitrogen for three times, an anhydrous THF (20 mL) and triethylamine (10 mL) were added into the Schlenk flask. Then the mixture was stirred at 70 °C for 4 days. After cooling to room temperature, the mixture was extracted with CHCl_3 , and the crude product was purified by flash column chromatography with CHCl_3 : ethanol (100: 3, v/v) and the ligand **LB** were obtained (0.27 g, 65% yield) as a yellow solid. ^1H NMR conditions: (500 MHz, CDCl_3 , 300 K); characterization, δ (ppm): 8.72 (s, 8H, $\text{tpy-H}^{3,5'}$), 8.70 (d, $J = 4.5$ Hz, 8H, $\text{tpy-H}^{6,6''}$), 8.67 (d, $J = 7.5$ Hz, 8H, $\text{tpy-H}^{3,3''}$), 7.87 (td, $J = 7.7$, 1.9 Hz, 8H, $\text{tpy-H}^{4,4''}$), 7.77 (d, $J = 2.2$ Hz, 4H, H^b), 7.53 (dd, $J = 8.5$, 2.1 Hz, 4H, H^b), 7.37–7.32 (m, 16H, $\text{tpy-H}^{5,5''}$ and H^d), 7.06 (d, $J = 8.1$ Hz, 8H, H^e), 6.98 (d, $J = 8.7$ Hz, 4H, H^c), 4.06 (t, $J = 6.2$ Hz, 8H, H^f), 1.74 (m, 8H), 1.45–1.39 (m, 8H), 1.16 (m, 16H), 0.75 (t, $J = 7.0$ Hz, 12H). ^{13}C NMR conditions: (125 MHz, CDCl_3 , 300 K); characterization, δ (ppm): 156.45, 155.17, 149.12, 147.58, 142.88, 140.90, 136.68, 133.86, 133.33, 131.47, 131.07, 128.47, 123.55, 121.97, 121.81, 121.17, 115.59, 112.15, 89.78, 88.67, 68.64, 31.53, 29.08, 25.78, 22.36, 13.93. MALDI-TOF MS characterization, (m/z): Calcd for $[\text{C}_{142}\text{H}_{120}\text{N}_{12}\text{O}_4\text{-e}]^+$ 2056.96. Found: 2056.95.

Synthesis of cage $[\text{Zn}_6\text{LA}_3]$

The solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (2.3 mg, 7.6 μmol) dissolved in MeOH (3 mL) was accurately added into another solution of ligand **LA** (7.5 mg, 3.8 μmol) in CHCl_3 (1.0 mL), then the mixture was heated at 50 °C for 10 h. After cooling to room temperature, 40 mg NH_4PF_6 was added and bright yellow precipitate was observed, washed twice with water, and a yellow solid product was obtained in 91% yield. ^1H NMR conditions: (500 MHz, CD_3CN , 300 K); characterization, δ (ppm): 9.02 (s, 8H,

$\text{tpy-H}^{3,5'}$), 8.52 (d, $J = 8.2$ Hz, 8H, $\text{tpy-H}^{3,3''}$), 8.16 (s, 4H, H^b), 7.97 (d, $J = 8.8$ Hz, 4H, H^b), 7.83 (t, $J = 8.0$ Hz, 8H, $\text{tpy-H}^{4,4''}$), 7.75 (m, 16H, $\text{tpy-H}^{6,6''}$ and H^d), 7.40 (m, 12H, H^c and H^e), 6.97 (t, $J = 6.5$ Hz, 8H, $\text{tpy-H}^{5,5''}$), 4.28 (s, 8H, H^f), 1.92–1.84 (m, 8H), 1.50 (m, 8H), 1.29–1.19 (m, 8H), 1.05 (m, 8H), 0.57 (t, $J = 7.4$ Hz, 12H). ^{13}C NMR conditions: (125 MHz, CD_3CN , 300 K); characterization, δ (ppm): 156.14, 154.60, 148.78, 147.89, 147.79, 142.46, 140.91, 137.54, 137.42, 132.88, 131.33, 130.50, 130.09, 128.89, 127.06, 125.90, 125.83, 124.30, 122.98, 113.34, 68.90, 31.35, 29.07, 25.81, 22.21, 13.13. ESI-MS characterization: (m/z): 1458.9 $[\text{M-5PF}_6]^{5+}$ (calcd m/z : 1458.9), 1191.6 $[\text{M-6PF}_6]^{6+}$ (calcd m/z : 1191.6), 1000.7 $[\text{M-7PF}_6]^{7+}$ (calcd m/z : 1000.7), 857.5 $[\text{M-8PF}_6]^{8+}$ (calcd m/z : 857.5), 746.1 $[\text{M-9PF}_6]^{9+}$ (calcd m/z : 746.1) and 657.0 $[\text{M-10PF}_6]^{10+}$ (calcd m/z : 657.0).

Synthesis of cage $[\text{Zn}_6\text{LB}_3]$

The solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (2.1 mg, 7.0 μmol) in MeOH (3 mL) was added accurately into another solution of ligand **LB** (7.2 mg, 3.5 μmol) in CHCl_3 (1.0 mL), then the mixture was heated at 50 °C for 10 h. After cooling to room temperature, 35 mg NH_4PF_6 was added and bright yellow precipitate was observed. The product was washed twice with water, and a yellow solid was obtained in 90% yield. ^1H NMR conditions: (500 MHz, CD_3CN , 300 K); characterization, δ (ppm): 9.00 (s, 8H, $\text{tpy-H}^{3,5'}$), 8.58 (d, $J = 8.6$ Hz, 8H, $\text{tpy-H}^{3,3''}$), 8.15 (s, 4H, H^b), 8.10–8.02 (m, 8H, $\text{tpy-H}^{4,4''}$), 7.87–7.75 (m, 12H, $\text{tpy-H}^{6,6''}$ and H^b), 7.43 (s, 8H, H^d), 7.39–7.25 (m, 12H, $\text{tpy-H}^{5,5''}$ and H^c), 7.18 (d, $J = 8.2$ Hz, 8H, H^e), 4.28 (s, 8H, H^f), 1.88 (s, 8H), 1.49 (s, 8H), 1.26 (s, 8H), 1.08 (m, 8H), 0.70–0.52 (m, 12H). ^{13}C NMR conditions: (125 MHz, CD_3CN , 300 K); characterization, δ (ppm): 156.72, 153.74, 149.05, 147.99, 147.89, 141.23, 131.32, 130.85, 127.46, 127.36, 125.79, 124.23, 124.13, 122.98, 121.62, 115.67, 113.36, 89.15, 88.65, 69.09, 31.31, 28.93, 25.75, 22.22, 13.15. ESI-MS characterization: (m/z): 1516.6 $[\text{M-5PF}_6]^{5+}$ (calcd m/z : 1516.6), 1239.7 $[\text{M-6PF}_6]^{6+}$ (calcd m/z : 1239.7), 1041.8 $[\text{M-7PF}_6]^{7+}$ (calcd m/z : 1041.8), 893.5 $[\text{M-8PF}_6]^{8+}$ (calcd m/z : 893.5), 778.1 $[\text{M-9PF}_6]^{9+}$ (calcd m/z : 778.1), 685.8 $[\text{M-10PF}_6]^{10+}$ (calcd m/z : 685.8), and 610.3 $[\text{M-11PF}_6]^{11+}$ (calcd m/z : 610.3).

Results and Discussion

Preparation and characterization of ligands and supramolecules

In coordination-driven self-assembly, tpy has been widely used as powerful building block to construct rigid supramolecular architectures.^{63–66} In this study, tpy-based ligands, **LA** and **LB**, were prepared through Suzuki and Sonogashira couplings, as shown in Figure 1. Both **LA** and **LB** were designed with TPE core to afford strong emission properties. By contrast with **LA**, **LB** contained additional alkyne connectivity to increase the rigidity

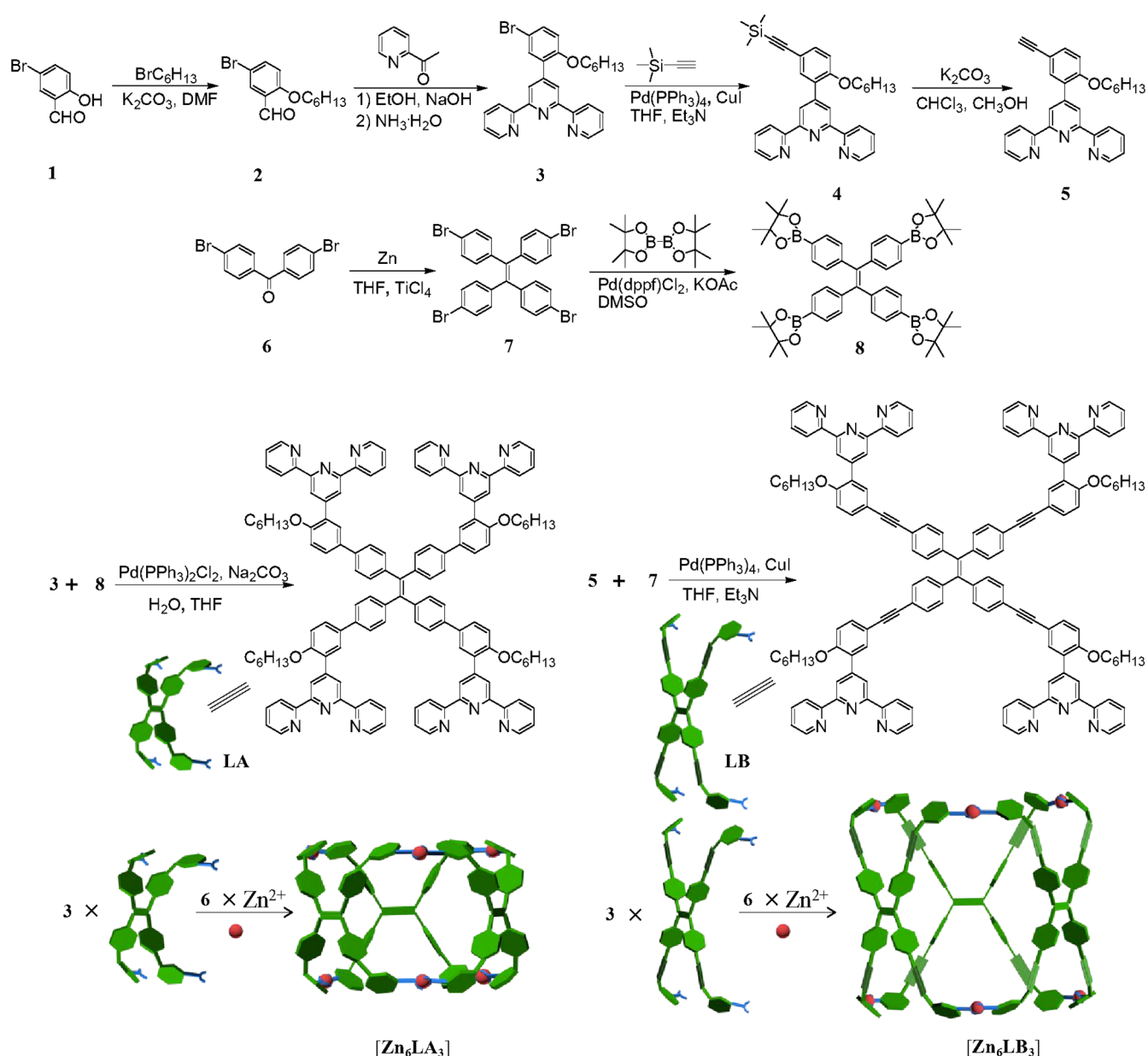


Figure 1 | Synthesis of the ligands **LA**, **LB**, and self-assembly of the complexes **[Zn₆LA₃]** and **[Zn₆LB₃]**.

of the structure. These two ligands were assembled with $\text{Zn}(\text{NO}_3)_2\cdot 6\text{H}_2\text{O}$ in a precise 1:2 stoichiometric ratio in a solvent mixture of $\text{CHCl}_3/\text{MeOH}$ at 50 °C for 10 h, followed by addition of an excess ammonium hexafluorophosphate (NH_4PF_6) salt to give yellow precipitate products (yields: 91% for **[Zn₆LA₃]** and 90% for **[Zn₆LB₃]**).

These ligands and complexes were characterized by ^1H NMR (Supporting Information Figures S1, S5, S9, and S14), ^{13}C NMR (Supporting Information Figures S2, S6, S10, and S15), 2D correlation spectroscopy (2D-CO-SY) (Supporting Information Figures S3, S4, S7, S8, S11, S12, S16, and S17), 2D diffusion-ordered NMR spectroscopy (2D-DOSY) (Supporting Information Figures S13 and S18), matrix-assisted laser desorption/ionization

time-of-flight (MALDI-TOF) (Supporting Information Figures S19 and S20), ESI-MS (Supporting Information Figures S21 and S22) and TWIM-MS. First, the ^1H NMR spectroscopy documented the formation of highly symmetrical metallacages (Figure 2). Compared with their respective ligands, the cages, **[Zn₆LA₃]** and **[Zn₆LB₃]**, displayed broadened peaks of all the protons, suggesting the formation of large complexes due to the slow tumbling motion⁶⁷ on the NMR time scale. 3',5' protons of **[Zn₆LA₃]** shifted downfield (~ 0.29 ppm), compared with the signal of **LA**, while the protons at 6,6" position of tpy were significantly shifted upfield (~ 0.91 ppm) due to the electron shielding effect⁶⁸ (Figure 2a). For **[Zn₆LB₃]**, the signals of 3',5' protons shifted downfield (~ 0.28 ppm), and 6,6" protons shifted

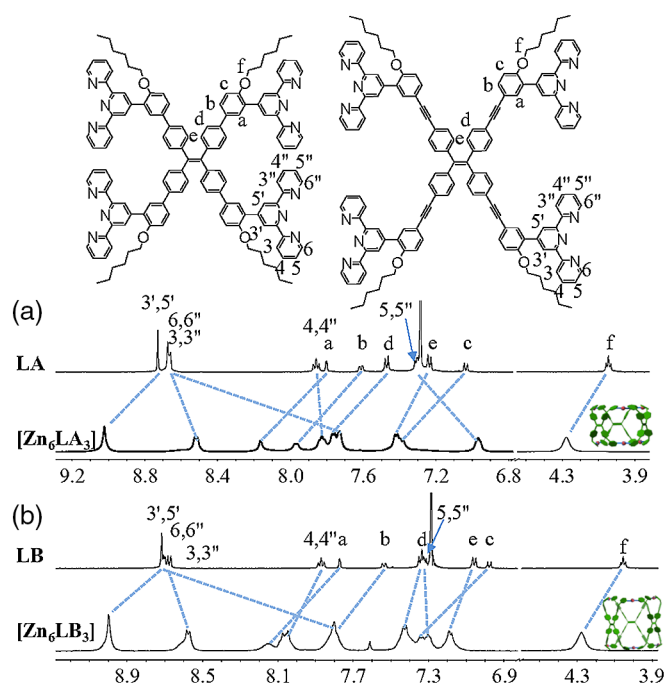


Figure 2 | ^1H NMR spectra (500 MHz). (a) **LA** in CDCl_3 and cage $[\text{Zn}_6\text{LA}_3]$ in CD_3CN . (b) **LB** in CDCl_3 and cage $[\text{Zn}_6\text{LB}_3]$ in CD_3CN .

upfield (~ 0.95 ppm) (Figure 2b). Collectively, such NMR results suggested the formation of discrete metalla-supramolecules rather than the random polymers attained by tpy-Zn(II) coordination. Second, molecular simulations using the Materials Studio software showed that the diameters of complexes $[\text{Zn}_6\text{LA}_3]$ and $[\text{Zn}_6\text{LB}_3]$ were 2.9 and 3.6 nm, respectively. To prove the size of the simulated structures, we measured the real molecular size by DOSY. The signal bands of complexes $[\text{Zn}_6\text{LA}_3]$ and $[\text{Zn}_6\text{LB}_3]$ were observed at the logarithm of diffusion coefficient ($\log D$) = -9.34 and -9.48 , respectively. Accordingly, the experimental diameters of $[\text{Zn}_6\text{LA}_3]$ and $[\text{Zn}_6\text{LB}_3]$, calculated via the Stokes-Einstein equation, were 2.7 and 3.6 nm, which proved the formation of simulated discrete supramolecular structures (Supporting Information Figures S13 and S18).

ESI-MS and TWIM-MS were used to validate further the molecular compositions of the metallacages.^{63,69} For cage $[\text{Zn}_6\text{LA}_3]$, only one set of peaks with different charge states (5+ to 10+) were observed with successive loss of the corresponding PF_6^- counterions (Figure 3a). The experimental molecular weight (MW) obtained was 8006.0 Da, which matched well with the formula of $[(\text{C}_{134}\text{H}_{120}\text{N}_{12}\text{O}_4)_3\text{Zn}_6(\text{PF}_6)_{12}]$. The experimental isotope pattern of each charge state was consistent with the theoretical simulation (Supporting Information Figure S21). Moreover, all the different charge states revealed by TWIM-MS spectrum had narrow drift time distribution, which indicated that no overlapping isomers

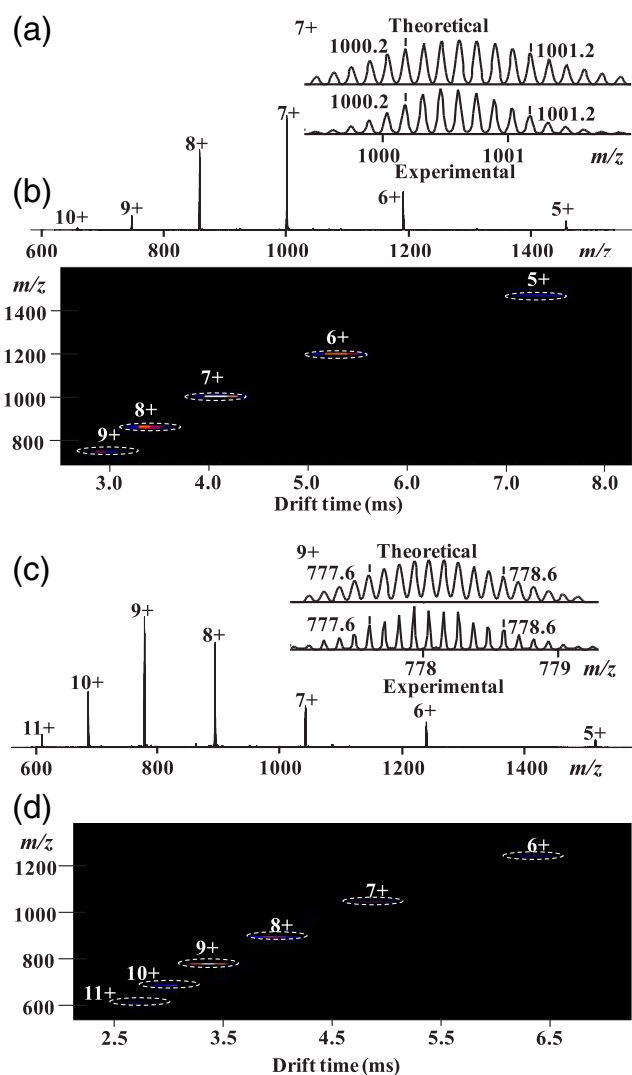


Figure 3 | (a) ESI-MS and (b) 2D TWIM-MS plot of $[\text{Zn}_6\text{LA}_3]$. (c) ESI-MS and (d) 2D TWIM-MS plot of $[\text{Zn}_6\text{LB}_3]$.

or structural conformers were generated (Figure 3b).^{63,69–76} For cage $[\text{Zn}_6\text{LB}_3]$, the experimental MW obtained was 8294.0 Da, corresponding to the formula of $[(\text{C}_{142}\text{H}_{120}\text{N}_{12}\text{O}_4)_3\text{Zn}_6(\text{PF}_6)_{12}]$. Also, there was only one prominent set of signals for the multicharged entities from 5+ to 11+ shown in the ESI-MS spectrum (Figure 3c). Each peak was in an excellent agreement with the corresponding theoretical isotope pattern (Supporting Information Figure S22). Moreover, the cage $[\text{Zn}_6\text{LB}_3]$ displayed a narrowly distributed band of signals in TWIM-MS, which verified the successful formation of a discrete metallacage (Figure 3d).

Photophysical studies of ligands and supramolecules in solution

With the well-defined supramolecular cages in hand, we systematically characterized the ligands and

supramolecules by UV-vis absorption and fluorescence spectroscopy. By comparing the absorption spectra of **LA** and $[\text{Zn}_6\text{LA}_3]$, the absorption band of **LA** (300 nm) originated mainly from the superposition of TPE and tpy. Compared to **LA**, $[\text{Zn}_6\text{LA}_3]$ had an enhanced absorption peak at ~ 300 nm, which might have originated from the metal-to-ligand charge transfer (MLCT) after coordination (Figure 4a). In the fluorescence spectrum of **LA**, the emission band peaked at ~ 425 nm, ascribed to the tpy unit⁷⁷; however, the emission of TPE unit is not observed (Figure 4a), suggesting that the rotation of TPE in **LA** could not be inhibited effectively. The quantum efficiency of fluorescence of **LA** is 13.8%. Nonetheless, the fluorescence of the prepared supramolecular cage $[\text{Zn}_6\text{LA}_3]$ redshifted significantly with an emission peak at ~ 510 nm (Figure 4a), attributable to the partially restricted rotation of TPE by coordination. Thus the changes of the molecular conformation led to an intramolecular charge transfer process, resulting in the redshift of the emission wavelength.⁷⁸ It is worth noting that, although there was a strong emission of tpy in the ligand, after the ligand was coordinated with the metal, its excited-state energy was transferred to the metal center through MLCT and rapid deactivation via nonradiative channels, which resulted in

fluorescence quenching.^{77,79} The quantum efficiency of $[\text{Zn}_6\text{LA}_3]$ reached only 4.67%, even though it had a strong MLCT effect.

Moreover, by introducing additional alkyne connectivity into the building blocks, the optical spectra of **LB** and $[\text{Zn}_6\text{LB}_3]$ (Figure 4b) were completely different from the **LA** series shown in Figure 4a. The absorptions of **LB** and $[\text{Zn}_6\text{LB}_3]$ were attributed mainly to the TPE, tpy, and MLCT state in the UV-vis spectra, in which the absorption intensity of TPE moiety at ~ 320 nm was significantly enhanced. In the fluorescence spectrum of **LB** and $[\text{Zn}_6\text{LB}_3]$ (Figure 4b), the emission of **LB** consists of two bands: one located at ~ 430 nm originated from the tpy unit, and the other peak at ~ 550 nm is attributable to TPE unit. The quantum efficiency of fluorescence increased to 28.0%. This could be due to the introduction of the alkyne bond, which increased the rigidity and extended the conjugation, thus reducing the twisted conformation of the molecule, and resulting in enhanced emission.^{80,81} Similar to $[\text{Zn}_6\text{LA}_3]$, the emission peak of $[\text{Zn}_6\text{LB}_3]$ at ~ 510 nm is attributable to TPE. In contrast, the emission of the tpy moiety is quenched by MLCT after coordination with Zn(II). Thus, the quantum efficiency of $[\text{Zn}_6\text{LB}_3]$ is 20.79%, and 4.5 times as high as that of $[\text{Zn}_6\text{LA}_3]$.

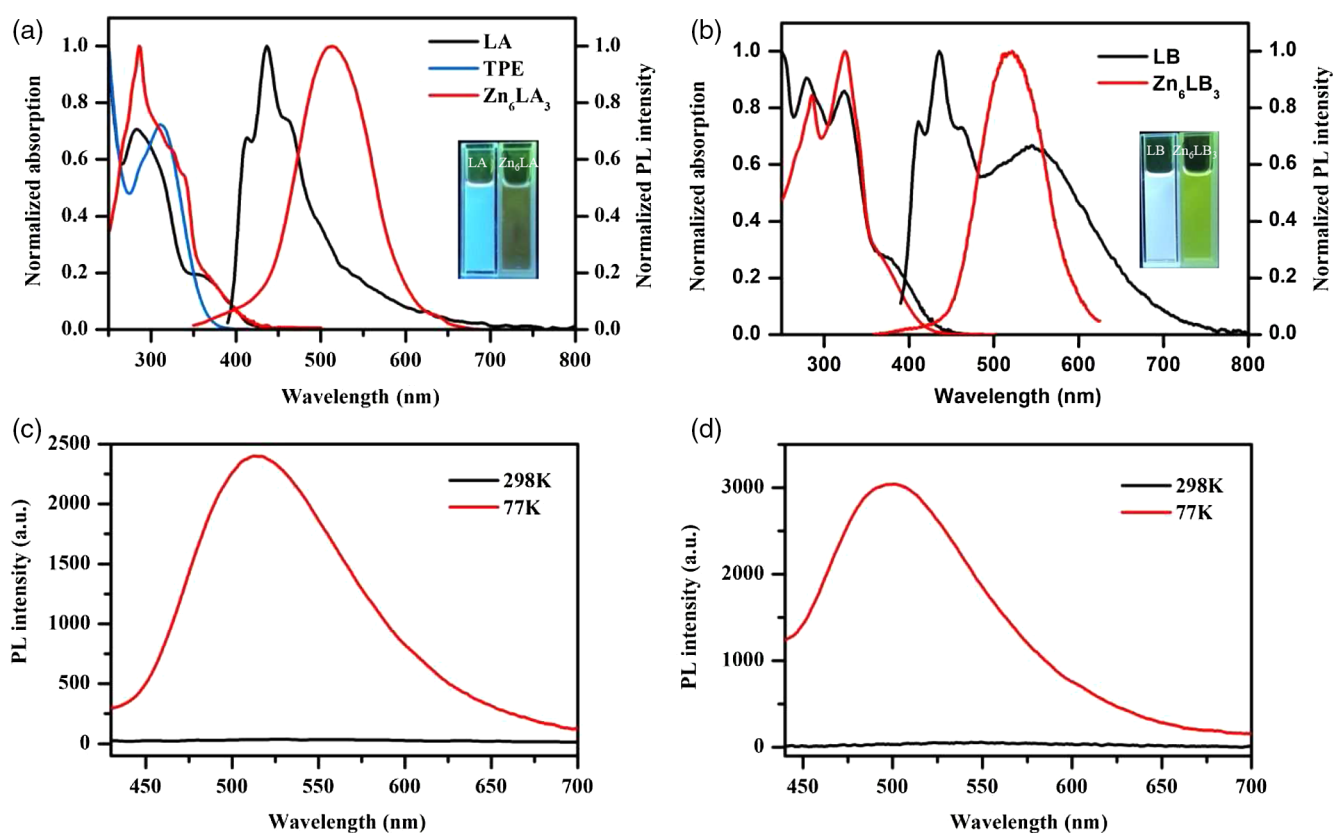


Figure 4 | (a) Normalized UV-visible and photoluminescence (PL) spectra of TPE, **LA**, and $[\text{Zn}_6\text{LA}_3]$ in acetonitrile ($\lambda_{\text{ex}} = 320$ nm, $c = 1.00$ μM). (b) Normalized UV-Vis and PL spectra of **LB** and $[\text{Zn}_6\text{LB}_3]$ in acetonitrile ($\lambda_{\text{ex}} = 320$ nm, $c = 1.00$ μM). Low-temperature PL spectra of (c) $[\text{Zn}_6\text{LA}_3]$ and (d) $[\text{Zn}_6\text{LB}_3]$ in acetonitrile ($\lambda_{\text{ex}} = 365$ nm, $c = 1.00$ μM).

As such, a subtle change in shift occurred by introducing an alkyne bond into the ligand, likely to increase the emission of supramolecular cages in dilute solution significantly. Besides, we performed the low-temperature fluorescence spectra on the two cages by freezing these supramolecules rapidly in acetonitrile solution to reach a glassy state, using liquid nitrogen at 77 K. This state limited the intramolecular rotation of TPE profoundly and led to a significant increase in the fluorescence intensity (Figure 4c and 4d).

Further, we studied the fluorescence property of the hybrid supramolecular cages by mixing **LA** and **LB** at varying ratios for self-assembly. When **LA** and **LB** were assembled in a ratio of 1:1 with the Zn(II) metal, the ESI-MS data exhibited the formation of a series of hybrid supramolecular cages rather than individual $[\text{Zn}_6\text{LA}_3]$ or $[\text{Zn}_6\text{LB}_3]$ via self-sorting. Such a statistical distribution of the mixtures included four supramolecules, viz, $[\text{Zn}_6\text{LA}_3]$, $[\text{Zn}_6\text{LA}_2\text{LB}]$, $[\text{Zn}_6\text{LALB}_2]$, $[\text{Zn}_6\text{LB}_3]$ (Figure 5 and Supporting Information Figures S23–S26). As shown in the fluorescence spectra of assemblies with different ratios of **LA** and **LB** (Figure 6a), the quantum efficiencies of these hybrid systems increased proportionally with the percentage of **LB** (Figure 6b). These results suggest that the ligand **LB**, which has a higher degree of conjugation,

is more favorable for the emission of TPE than **LA**, and this behavior did not change, even in an instance of fabrication of the hybrid cages.

Photophysical studies of ligands and supramolecules in the solid-state

Finally, we sought to address whether the emission of AIE-based ligands and the supramolecules would respond to external hydrostatic pressure in the solid-state. The photomicrograph and spectra in the solid state were collected under varying hydrostatic pressure conditions, as shown in Supporting Information Figure S27 and Figure 7. Under high pressure (Supporting Information Figure S27), the emission wavelength of **LA** redshifted from ~ 530 nm (0 GPa) to ~ 600 nm (4.65 GPa), and the emission wavelength of $[\text{Zn}_6\text{LA}_3]$ also redshifted from ~ 540 nm (0 GPa) to ~ 580 nm (2.88 GPa) (Supporting Information Figures S27a and S27d). Meanwhile, the emission intensity of **LA** and $[\text{Zn}_6\text{LA}_3]$ decreased at increasing pressure. This could be due to the tighter molecular packing that enhanced the π - π and dipole-dipole interactions under

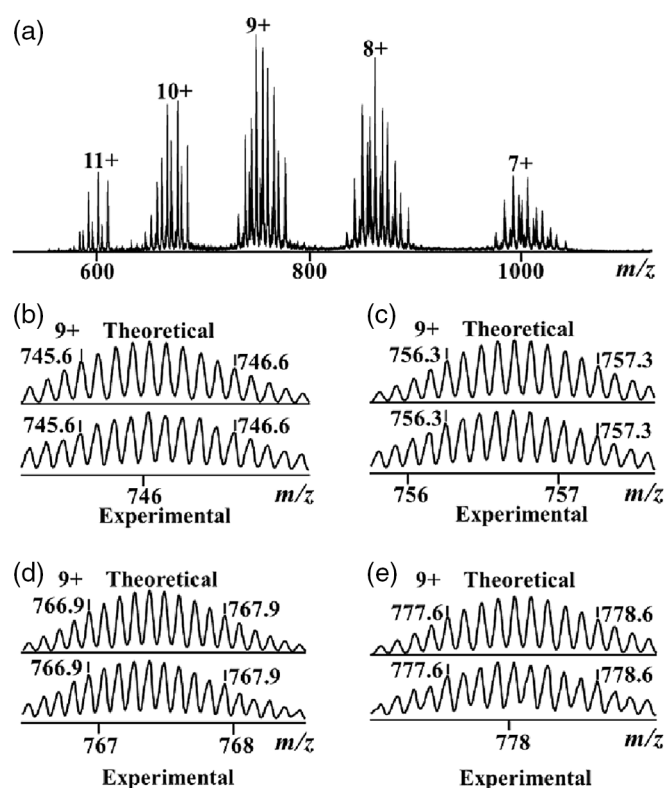


Figure 5 | (a) ESI-MS of self-assembly of **LA** and **LB** in a ratio of 1:1. ESI-MS experimental data and theoretical isotope distribution of (b) $[\text{Zn}_6\text{LA}_3-9\text{PF}_6]^{9+}$, (c) $[\text{Zn}_6\text{LA}_2\text{LB}-9\text{PF}_6]^{9+}$, (d) $[\text{Zn}_6\text{LALB}_2-9\text{PF}_6]^{9+}$, (e) $[\text{Zn}_6\text{LB}_3-9\text{PF}_6]^{9+}$.

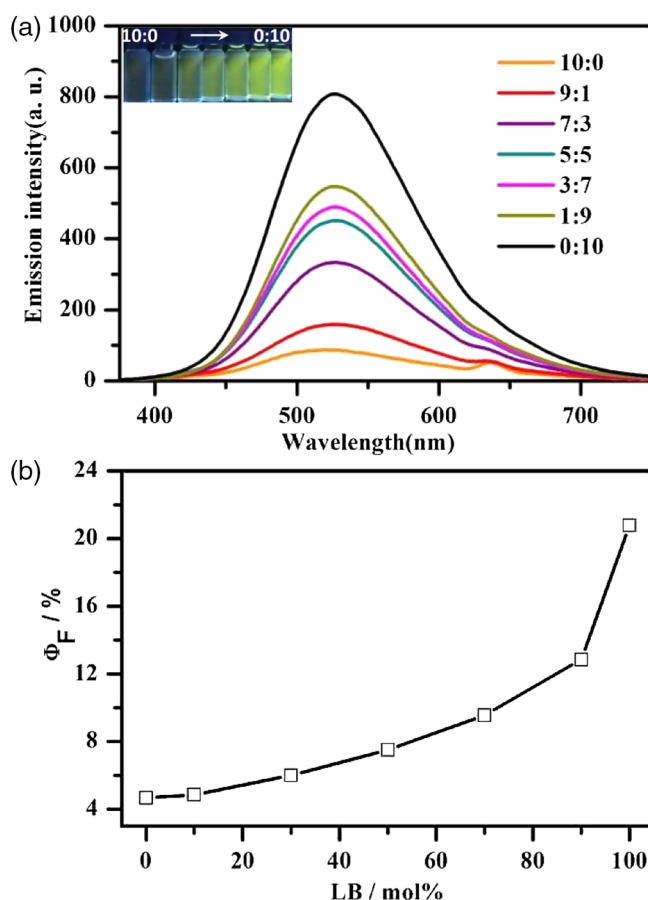


Figure 6 | (a) Photoluminescence (PL) spectra and (b) quantum yields in acetonitrile of ligands **LA**, **LB** assembled in different ratios ($\lambda_{\text{ex}} = 320$ nm, $c = 1.00$ μM).

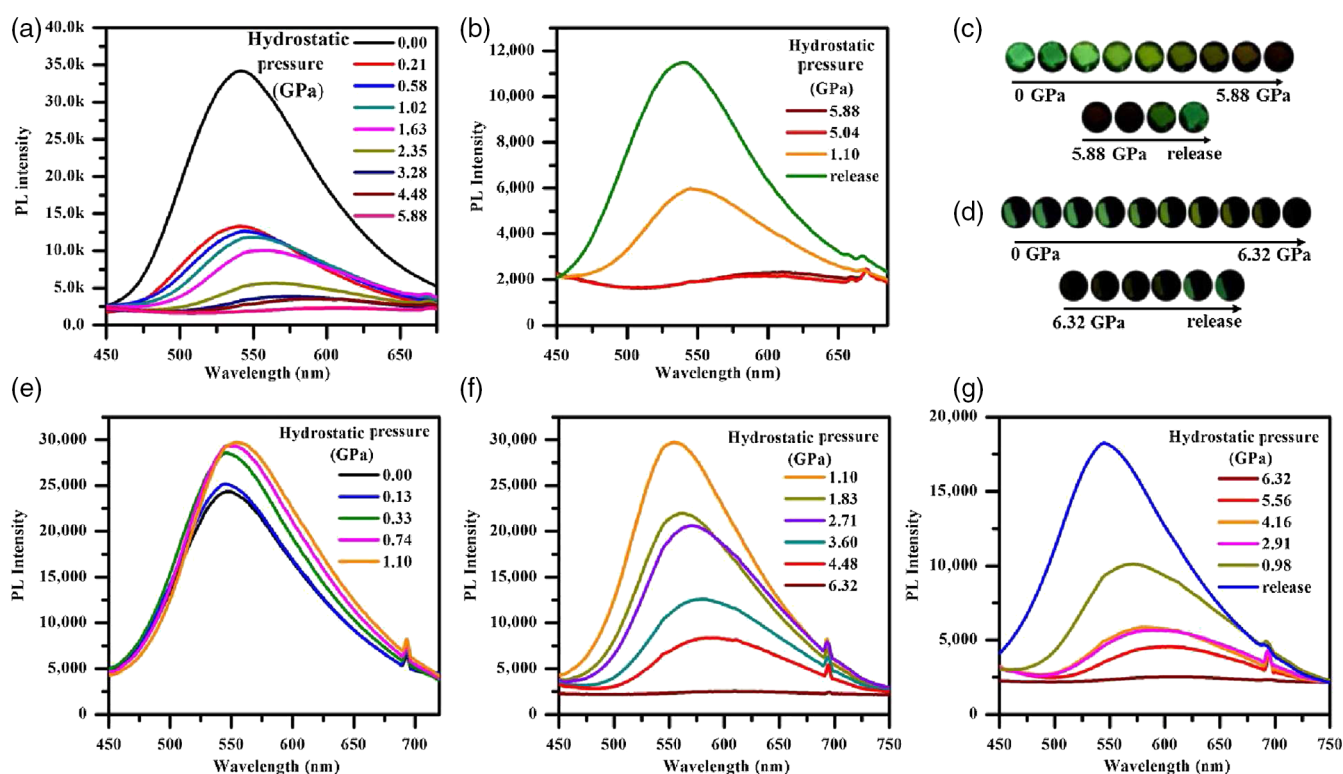


Figure 7 | Effects of applied pressure on AIE-based ligand and their supramolecular derivatives. Fluorescence emission responses for (a) **LB**, at varying applied pressure, (b) **LB**, gradual pressure withdrawal. Photomicrographs of (c) **LB** (solid) and (d) **[Zn₆LB₃]** (solid). Fluorescence emission responses upon for (e) **[Zn₆LB₃]**, varying applied pressure, (f) **[Zn₆LB₃]**, varying applied pressure, and (g) **[Zn₆LB₃]**, gradual pressure withdrawal. ($\lambda_{\text{ex}} = 365 \text{ nm}$).

high pressure with the consequent fluorescence quenching.^{82,83} Moreover, the fluorescence intensity of **LA** and **[Zn₆LA₃]** recovered gradually with the progressive removal of the applied pressure (Supporting Information Figures S27b and S27e).

Similarly, for ligand **LB**, the emission intensity was continuously reduced as pressure increased from 0 to 5.88 GPa (Figure 7a), also demonstrated as representative photomicrograph images in Figure 7c. We also ascribed this observation to the excessive atomic packing leading to the quenched fluorescent emission in this process. More interestingly, the fluorescence intensity of **[Zn₆LB₃]** increased significantly under low hydrostatic pressure from 0 to 1.10 GPa (Figure 7e). However, when the pressure was increased gradually above 1.10 GPa to reach 6.32 GPa eventually, during which the fluorescence began to quench (Figures 7f), which is also displayed as representative photomicrograph images in Figure 7d. However, when the pressure was increased gradually above 1.10 GPa to reach 6.32 GPa eventually, during which the fluorescence began to quench (Figures 7f). Thus, for **[Zn₆LB₃]**, increasing the pressure might have caused further limitation on the rotation of TPE moieties leading to the energy loss through radiative transitions

with the subsequent increase in emission efficiency at applied pressures ranging from 0 to 1.10 GPa. On the other hand, when the pressure continued to increase until it reached 6.32 GPa, excessive molecular packing might have occurred, which, to some extent, ultimately quenched the excited state. Under normal pressure, the maximum emission of **LB** and **[Zn₆LB₃]** was at ~ 540 and 560 nm, respectively. Upon compression, the corresponding fluorescence of **LB** and **[Zn₆LB₃]** redshifted to ~ 610 nm at 5.88 GPa and ~ 620 nm at 6.32 GPa. When the pressure was withdrawn gradually, the fluorescence of both **LB** and **[Zn₆LB₃]** could recover to an intermediate level, and the corresponding emission wavelength was blueshifted to the initial state (Figure 7b and 7g).

Conclusion

We have prepared two TPE-based metallacages through the coordination-driven self-assembly of tpy ligands and Zn(II) successfully. Through introducing additional alkyne connectivity, based on **LA**, the elongated ligand **LB** with a high degree of rigidity and conjugation was able to assemble into a highly emissive cage **[Zn₆LB₃]** in dilute solution. The quantum efficiencies of these two

complexes were 4.67% for $[\text{Zn}_6\text{LA}_3]$ and 20.79% for $[\text{Zn}_6\text{LB}_3]$. These results enabled us to understand the luminescence behavior of metalla-supramolecules with AIE-active motifs, which guided our design of emissive supramolecules with desired properties. Further, we studied the emission properties of both ligands and their respective supramolecules in the solid-state by gradually increasing the external pressure. Through subtle adjustment of these structures, we found success in developing supramolecular cages with enhanced emission and stimulus-responsive luminescence features, thereby, enriching our insight into supramolecular chemistry involving AIE and the development of luminescent materials.

Supporting Information

Supporting Information is available.

Conflict of Interest

The authors declare no conflict of interest.

Acknowledgments

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